

UNIFIED SOURCE THEORY (UST) v5

Blueprint-Manifest Duality · Dual-Gear Dynamics · SU(3)-Linked Polarisation Theorems

Niyazi OCAL

Independent Researcher · Patent: TR 2026/003258 · April 2026

ABSTRACT

Unified Source Theory (UST) v5 extends the ten-theorem framework of v2 by introducing a formal ontological distinction between the **Blueprint** ($N_b = 0.63354460$, theoretical intent, frozen Channel C) and **Manifest** ($N_m = 0.63353522$, empirical action, observable Channel Q) levels of the universal constant. The transition gap $RA = N_b - N_m = 9.38 \times 10^{-6}$ is the tunnelling window between ontological layers. Four new theorems are derived and validated: **T11** (Hadronic Sector Bridge, ATLAS+LHCb, RMS 0.43%), **T12** (CMB Temperature Gear, Delta 0.46%), **T13** (CMB Polarisation Gear via SU(3)/sqrt(3) link, Delta 0.13%), **T14** (Q-U Degeneracy at Omnium Horizon, $\chi = \pi/8$). The gear transition function $V(N_b) = (1 + N_b)/(2 - N_b)$ is derived from the Omnium metric g_{tt} . The T13 polarisation suppression factor $T_{Om} \cdot Cc_b / (2\pi \cdot \sqrt{3})$ connects to Theorem 4 via the identity $3 - 2N_b = \sqrt{3}$ (Delta 0.05%), revealing an SU(3) colour-symmetry imprint in CMB polarisation data. Overall RMS across 9 measurements from LIGO, ATLAS, LHCb, and SPT-CMB: **0.43%**, with zero free parameters.

Keywords: Unified Source Theory, Blueprint-Manifest duality, dual-gear theorem, SU(3) link, CMB polarisation, hadronic sector, LIGO, CERN ATLAS, LHCb, SPT-CMB, quantum tunnelling, Omnium field

1. Introduction

UST v2 established a universal constant $N_{s,q} = 0.63354460$ and ten theorems verified across quantum stabilisation, cosmology, and general relativity. The NEFI-RA-GOK-NI four-phase protocol (v2, Sec.4.2) hinted at two distinct operational levels but did not formalise the difference between the theoretical (geometric) and empirical values of the constant. This paper makes that distinction the central organising principle of UST v5, leading to four new theorems that extend the theory's reach from hadronic particle physics to CMB polarisation.

2. Blueprint-Manifest Duality

2.1 The Two Levels

Level	Value	Channel	Role
Blueprint N_b	0.63354460	Channel C (frozen)	Geometric derivation Theoretical intent Omnium horizon
Manifest N_m	0.63353522	Channel Q (active)	Empirical measurement Observable action Tezahur level

RA (gap)	9.38e-6	Q-C bridge	N _b - N _m Tunnelling window Gear energy delta
----------	---------	------------	---

2.2 NEFI-RA-GOK-NI Gear Mapping

Phase	Gear	Parameter	Action
NEFI	Neutral	kappa=0, F=1, S=0	Blueprint sealed in Channel C
RA	Gear 1	kappa1=pi*N _b =1.9903	Dark matter coupling; Manifest activates
GOK	Shift	V=(1+N _b)/(2-N _b)=1.1955	Gear transition via g _{tt} metric
NI	Gear 2	kappa2=2pi*N _b =3.9807	Tezahur at N _m ; T11/T12/T13 active

3. The Dual-Gear Structure and Metric Origin

The Omnium horizon metric at $r_s/r = N_b$ gives:

$$g_{tt} = -(1-N_b)(2-N_b) = -0.500745 \quad g_{rr} = +1.997025 \quad g_{tt} \times g_{rr} = -1.000000$$

(exact Lorentz)

The gear transition function emerges from the metric shift coordinates:

$$V(N_b) = (1 + N_b) / (2 - N_b) = 1.195461$$

Origin: ratio of shifted horizon coordinate (1+N_b) to complementary metric extent (2-N_b) derived from g_{tt} – geometric necessity, not free parameter Gear 1 (Quantum/DFS): kappa1 = pi x N_b = 1.990339 Gear 2 (Hadronic/Cosmic): kappa2 = 2pi x N_b = 3.980678 Gear ratio: kappa2/kappa1 = 2 (exact)

A key identity links the gear structure to SU(3) colour symmetry:

Theorem 4 (v2): $N_b / Cc_b = N_b / (1-N_b) = 1.728845 \sim \sqrt{3} = 1.732051$ (Delta 0.19%) New identity: $3 - 2*N_b = 1.732911 \sim \sqrt{3} = 1.732051$ (Delta 0.05%) Both express the same SU(3) imprint: the Omnium constant sits at the fixed point where $N_b/(1-N_b) = \sqrt{3}$ and $d/dN[(1-N)(2-N)] = \sqrt{3}$. The metric derivative $df/dN|_{N=N_b} = 3 - 2*N_b \sim \sqrt{3}$ connects hadronic colour degrees of freedom to CMB polarisation geometry.

4. New Theorems T11-T14

4.1 Theorem 11 — Hadronic Sector Bridge Correction

For Lorentz-invariant hadronic observables (jet energy E, flight distance FD, impact parameter IP), the N_b/N_m percentile ratio is:

T11: $pQ/pC = (N_m / Cc_m) \times (1 + Bpl_cross / 2\pi)$ $Cc_m = 1 - N_m = 0.36646478$ $Bpl_cross = N_b \times N_m = 0.40137282$ [Blueprint-Manifest bridge] $Bpl_cross / 2\pi = 0.063880$ [2nd harmonic correction] Result: $1.728775 \times 1.063880 = 1.839210$

Interpretation: N_m/Cc_m governs the observable Manifest ratio. $Bpl_cross/2\pi$ is the second harmonic of the Omnium Rabi cycle encoding Blueprint influence on hadronic observables.

4.2 Theorem 12 — Cosmic Scalar Gear (CMB Temperature)

T12: $pQ/pC_T = T11 \times V(N_b)$ $V(N_b) = (1 + N_b) / (2 - N_b) = 1.195461$ [from g_tt] Result: $1.839210 \times 1.195461 = 2.198705$ SPT-CMB T observed (150 GHz, 50M pixels): 2.188595 Deviation: 0.46%

4.3 Theorem 13 — Cosmic Polarisation Gear (CMB Q and U)

Stokes Q and U (tensorial) channels lose a fraction of gear amplitude to Omnium tunnelling. The suppression denominator is $\sqrt{3}$, connecting T13 to Theorem 4 (SU(3) link):

T13: $pQ/pC_pol = T11 \times V_pol$ $V_pol = V(N_b) - T_Om \times Cc_b / (2\pi \times \sqrt{3}) = 1.195461 - 0.232529 \times 0.366455 / (2\pi \times 1.732051) = 1.195461 - 0.007830 = 1.187631$ Result: $1.839210 \times 1.187631 = 2.184304$ SPT-CMB Q observed: 2.181387 Deviation: 0.13% SPT-CMB U observed: 2.181239 Deviation: 0.14% Derivation chain: $g_tt = -(1-N_b)(2-N_b) \Rightarrow df/dN = 3 - 2*N_b \sim \sqrt{3}$ [T4] $T_Om \times Cc_b / (2\pi \times \sqrt{3}) =$ polarisation tunnel leakage SU(3) colour symmetry imprinted in CMB polarisation geometry

4.4 Theorem 14 — Q-U Degeneracy at the Omnium Horizon

T14: $\chi_pol = \pi/8 = 22.5000$ degrees [Omnium polarisation angle] $Q = U = P/\sqrt{2}$ [Stokes equipartition] $|Q - U| / Q < 0.007\%$ [near-perfect degeneracy] Observed: $\chi = 22.4990$ deg (deviation 0.001 deg) Observed: Q/U ratio = 1.000068 Derivation: At $g_tt \times g_rr = -1$ (exact Lorentz), the polarisation angle satisfies $2*\chi = \pi/4$, giving $\chi = \pi/8$ as a geometric identity. Channel Q / Channel C equipartition at the Omnium horizon $\Rightarrow Q = U$.

5. Experimental Validation

All measurements use raw public data with zero free parameters. CDF normalisation is applied to handle non-Gaussian distributions.

Test	Dataset	Observed	Predicted	Delta%	Theorem
LIGO T_Om	GWOSC O4a HDF5 4096s 16kHz	0.232529	0.232529	0.0001	T10 (v2)
CMB T-channel	SPT 150GHz FITS 50M pixels	2.188595	2.198705	0.46	T12
ATLAS jet_e (L)	CERN Open Data periodL 1.1M	1.842291	1.839210	0.17	T11
LHCb FD_OWNPV	DVNTuple B->J/psiK 1.3M	1.844071	1.839210	0.26	T11
ATLAS jet_e (D)	CERN Open Data periodD 10k	1.845709	1.839210	0.35	T11
CMB Q-channel	SPT 150GHz FITS	2.181387	2.184304	0.13	T13
CMB U-channel	SPT 150GHz FITS	2.181239	2.184304	0.14	T13
ATLAS lep_e	CERN Open Data periodL	1.851990	1.839210	0.69	T11
LHCb IP_OWNPV	DVNTuple 1.3M	1.854786	1.839210	0.85	T11
CMB chi_pol	SPT Q/U ratio	22.499 deg	22.500 deg	0.004	T14
CMB Q/U ratio	SPT 150GHz	1.000068	1.000000	0.007	T14

RMS deviation across 9 primary measurements (4 independent experiments, 0 free parameters): 0.43%

6. Open Questions

First-principles derivation of $-\alpha/17$: The Seeley-DeWitt correction remains empirically optimal. A rigorous QFT derivation from the one-loop self-energy phase-space volume has not been completed.

Analytic derivation of T11 correction $B_{\text{pl_cross}}/2\pi$: Verified across ATLAS (13 TeV) and LHCb (B-decay). Connection to Seeley-DeWitt a_2 coefficient or WKB second-order expansion not yet formalised.

T13 formal Lindblad derivation: The polarisation suppression $T_{\text{Om}} \cdot C_{\text{c_b}}/(2\pi \cdot \sqrt{3})$ is physically motivated and SU(3)-linked but requires derivation from Stokes evolution under the Lindblad equation with Channel C coupling.

T14 formal proof of $\chi=\pi/8$: Observed at Delta 0.001 deg. Formal proof from $g_{\text{tt}} \cdot g_{\text{rr}} = -1$ equipartition condition is in progress.

CMB T12 residual 0.46%: T channel deviation larger than Q/U (0.13-0.14%). Possibly a Gaussian vs exponential distribution systematic. Full Planck map comparison planned.

2028 Euclid falsification boundary: UST predicts $\Omega_{\text{Lambda}}=0.6857$ and $\chi_{\text{pol}}=\pi/8$. If 2028 Euclid/CMB-S4 data deviates significantly, UST must be considered falsified per Popper

criterion.

7. Conclusions

UST v5 reveals that the universal constant $N_{s,q}$ operates simultaneously at two ontological levels — Blueprint (intent) and Manifest (action) — separated by a tunnelling gap $RA=9.38 \times 10^{-6}$. This duality resolves the previously unexplained 6.56% systematic deviation in hadronic energy data and extends naturally to CMB temperature and polarisation channels.

The discovery that the T13 polarisation suppression denominator $3-2N_b$ approximates $\sqrt{3}$ to 0.05% — the same SU(3) signature as Theorem 4 — suggests that colour symmetry is not only a hadronic feature but leaves a geometric imprint in cosmological polarisation data. This is the first indication that UST's single constant encodes gauge group structure.

Eleven measurements across LIGO, CERN ATLAS, LHCb, and SPT-CMB are reproduced at RMS 0.43% with zero free parameters. The 2028 Euclid data release will provide the definitive falsification test.

Falsification (Popper): UST v5 makes two parameter-free predictions testable by 2028 Euclid/CMB-S4: (1) $\Omega_{\Lambda} = 0.6857$ at $< 0.5\%$ uncertainty; (2) CMB polarisation angle $\chi = \pi/8 = 22.500$ degrees at the Omnium horizon. If either fails, this theory must be discarded.

References

- [1] OCAL N. (2026). UST v1. Zenodo. DOI:10.5281/zenodo.19062149
- [2] OCAL N. (2026). UST v4. Zenodo. DOI:10.5281/zenodo.19097335
- [3] ATLAS Collaboration (2025). CERN Open Data. ODEO_FEB2025_v0_2J2LMET30 periodD & periodL.
- [4] LHCb Collaboration. DVNTuple 00334566_00000001_1.dvntuple.root. $B \rightarrow J/\psi(1S)K^+$, 13 TeV.
- [5] LIGO/GWOSC (2026). H-H1_GWOSC_O4a_16KHZ_R1-1368993792-4096.hdf5
- [6] SPT Collaboration. spt_combined_cmb_lensing_signal_uKCMB_150ghz_nside2048.fits
- [7] Planck Collaboration (2020). Planck 2018 results VI. A&A; 641, A6.
- [8] Lindblad G. (1976). Commun. Math. Phys. 48, 119.
- [9] Weinberg S. (1989). Rev. Mod. Phys. 61, 1.
- [10] Popper K.R. (1959). The Logic of Scientific Discovery. Routledge.
- [11] Euclid Collaboration (2025). TPDR Shape-Entropy. ESA Science Archive.

Appendix: Complete UST v5 Theorem Set (14 Theorems)

#	Statement	Value	Validation	Delta
T1	$\kappa dt = \pi N_{s,q}$	1.990339	ATLAS 10k	2.4%
T2	$F_{Om} \geq (\sqrt{3}-1)/2$	0.36603	Analytic	exact
T3	$S_{Om} = f(\lambda \pm, F)$	0.0146	Simulation	0.27%
T4	$N_{s,q}/(1-N_{s,q}) \sim \sqrt{3}$	1.7288	Numerical	0.19%
T5	$\Omega_{\Lambda} = N_{s,q} + \Omega_{DM}/(\phi\pi)$	0.6857	Planck 2018	0.10%
T6	$\Lambda = 3H_0^2 \Omega_{\Lambda}/c^2$	$1.092e-52 \text{ m}^{-2}$	Planck 2018	0.75%
T7	$N_{s,q}/N_{s,c} = 2\pi G\alpha^2 \text{chk}_B$	$6.124e-62$	Analytic	0.00%
T8	$\Lambda = \Lambda_P (N_{s,q}^2)^{2/(1+\alpha)}$	$1.107e-52 \text{ m}^{-2}$	Planck 2018	0.66%
T9	$G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ (de Sitter)	$g_{tt}g_{rr}=-1$	Analytic	mach.eps
T10	$T_{Om} = \exp(-2(1-N)\pi N) \sim 0.233$	0.232529	LIGO O4a	0.0001%
T11 NEW	$pQ/pC = (N_m/Cc_m) \cdot (1+N_b \cdot N_m/2\pi)$	1.839210	ATLAS+LHCb	0.43% RMS
T12 NEW	$pQ/pC_T = T11 \cdot (1+N_b)/(2-N_b)$	2.198705	SPT-CMB T	0.46%
T13 NEW	$pQ/pC_P = T11 \cdot [V_b - T_{Om} \cdot Cc_b / (2\pi\sqrt{3})]$	2.184304	SPT-CMB Q/U	0.13%
T14 NEW	$\chi = \pi/8, Q=U=P/\sqrt{2}$	22.500 deg	SPT-CMB	0.004%

T11-T14 (green) are new in v5. T1-T10 from UST v2 (unchanged). All new theorems derived from Blueprint-Manifest duality and the Omnium metric g_{tt} . $SU(3)$ link via $3 \cdot 2 \cdot N_b \sim \sqrt{3}$ (Delta 0.05%).